Install library: `readxl`

# Time Series Regression

06

# Outline

- (10) Introduction
    - Time series (TS) linear regression: New Car Sales Example
- TS Forecasting with regression
    - (20) Excel
    - (20) R
    - (20) ex-ante vs ex-post forecasts

- (30) Regression assumptions: Diagnostic tests
- (15) Model selection
    - Useful predictors
- Appendix: Multiple regression-matrix notation

# Multiple linear **regression** model

- Previous classes: time series forecasting. Now, regression.

- $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_k x_{ki} + \epsilon_i$

where $k$ is the number of predictors.
  - If $k = 1$, the model is called a **simple** linear regression.
  - If $k \geq 2$, the model is called a **multiple** linear regression.
- $y_i$ **dependent** variable; **response** variable

- $x_{ki}$ the $i$th observation on the $k$th **independent variable**
  - $x_k$ is also called the **predictor** variable or the **explanatory variable**.

- Errors $\epsilon_i \sim NID(0, \sigma^2)$

# Time series regression Example

- **Dependent** variable
  (ML terms: **target** variable)
  - New Car Sales (NCS)
- **Independent** variables (**explanatory** variables)
  (ML terms: **predictors**, **features**)
  - Disposable personal income per capita (DPIPC)
  - Unemployment rate (UR)
  - Bank prime rate (PR)
  - University of Michigan Index of Consumer Sentiment (UMICS)

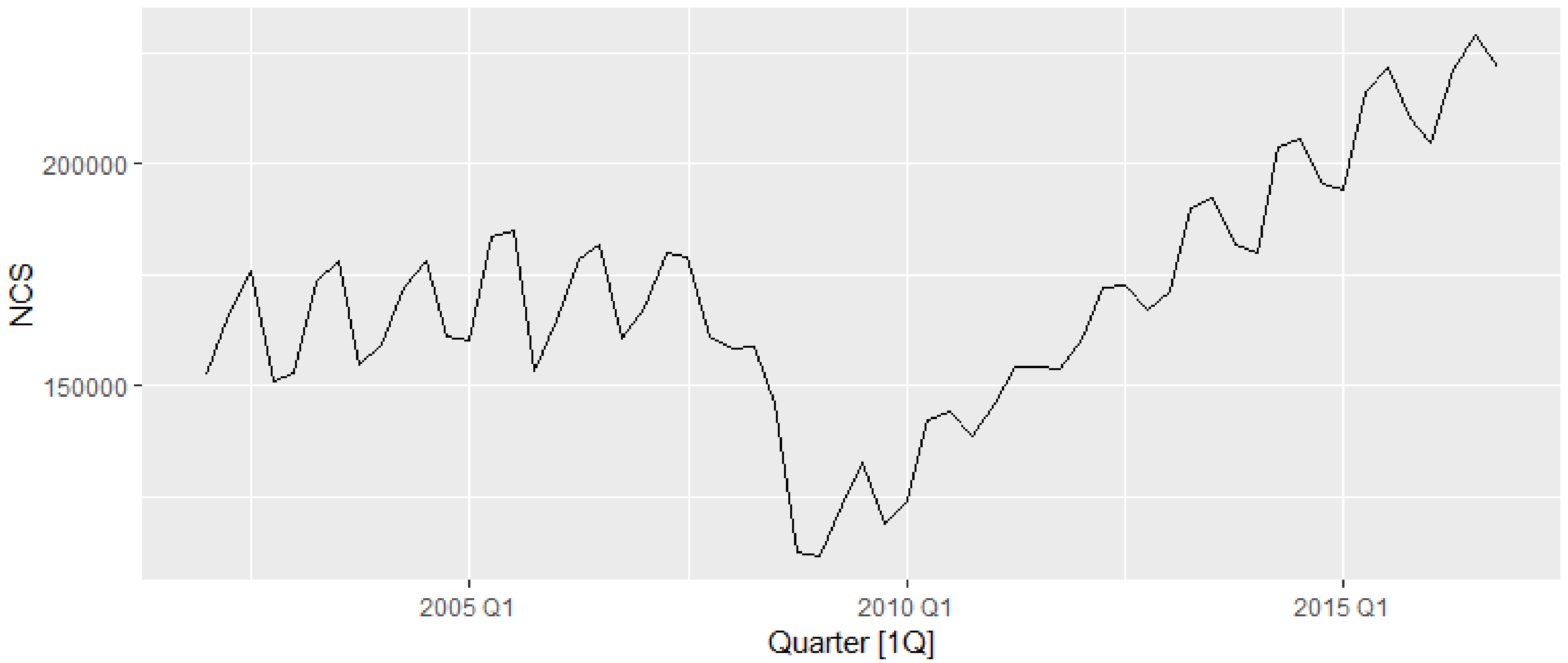
Sheet: `NewCarSales`

| Date | NCS | DPIPC | UR | PR | UMICS |
|---|---|---|---|---|---|
| Feb-02 | 152641 | 27840 | 5.7 | 4.75 | 93.13 |
| May-02 | 166036 | 28134 | 5.83 | 4.75 | 94.1 |
| Aug-02 | 175863 | 28168 | 5.73 | 4.75 | 87.27 |
| Nov-02 | 151219 | 28363 | 5.87 | 4.45 | 83.83 |
| … | | | | | |
| Feb-16 | 204681 | 42807 | 4.93 | 3.5 | 91.57 |
| May-16 | 221073 | 43265 | 4.87 | 3.5 | 92.4 |
| Aug-16 | 229123 | 43651 | 4.9 | 3.5 | 90.33 |
| Nov-16 | 221392 | 43759 | 4.7 | 3.55 | 93.07 |

The number of rows: $n = 60$.
How many years?

Positive or negative relationship?

# Example



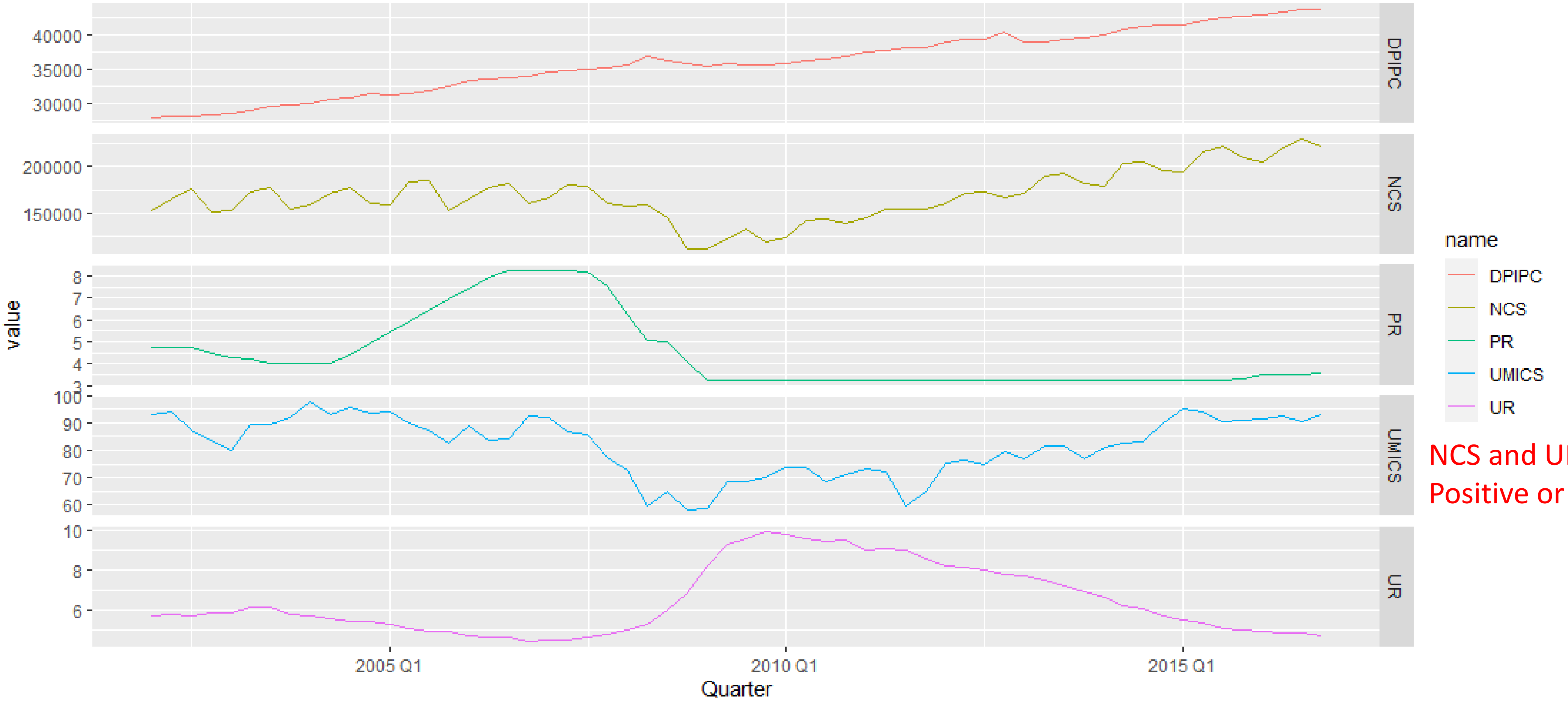

```
d <- read_excel("09-TS-regression.xlsx", sheet = "NewCarSales")

ncs <- d %>%
  mutate(Quarter = yearquarter("2002 Q1") + 0:59) %>%
  select(-Date) %>%
  relocate(Quarter) %>%
  as_tsibble(index = Quarter)

ncs %>% autoplot(NCS)
```

Sheet: `NewCarSales`

| Date | NCS | DPIPC | UR | PR | UMICS |
|---|---|---|---|---|---|
| Feb-02 | 152641 | 27840 | 5.7 | 4.75 | 93.13 |
| May-02 | 166036 | 28134 | 5.83 | 4.75 | 94.1 |
| Aug-02 | 175863 | 28168 | 5.73 | 4.75 | 87.27 |
| Nov-02 | 151219 | 28363 | 5.87 | 4.45 | 83.83 |
| … | | | | | |
| Feb-16 | 204681 | 42807 | 4.93 | 3.5 | 91.57 |
| May-16 | 221073 | 43265 | 4.87 | 3.5 | 92.4 |
| Aug-16 | 229123 | 43651 | 4.9 | 3.5 | 90.33 |
| Nov-16 | 221392 | 43759 | 4.7 | 3.55 | 93.07 |

```
> ncs
# A tsibble: 60 x 6 [1Q]
   Quarter     NCS DPIPC    UR    PR UMICS
     <qtr>   <dbl> <dbl> <dbl> <dbl> <dbl>
 1 2002 Q1 152641 27840  5.7   4.75  93.1
 2 2002 Q2 166036 28134  5.83  4.75  94.1
 3 2002 Q3 175863 28168  5.73  4.75  87.3
 4 2002 Q4 151219 28363  5.87  4.45  83.8
 5 2003 Q1 152843 28584  5.87  4.25  80.0
 6 2003 Q2 173360 28958  6.13  4.24  89.3
 7 2003 Q3 178086 29539  6.13  4     89.3
 8 2003 Q4 154916 29708  5.83  4     92.0
 9 2004 Q1 159086 30094  5.7   4     98
10 2004 Q2 172174 30537  5.6   4     93.3
# ... with 50 more rows
# i Use `print(n = ...)` to see more rows
```

# Example: **Time plot**

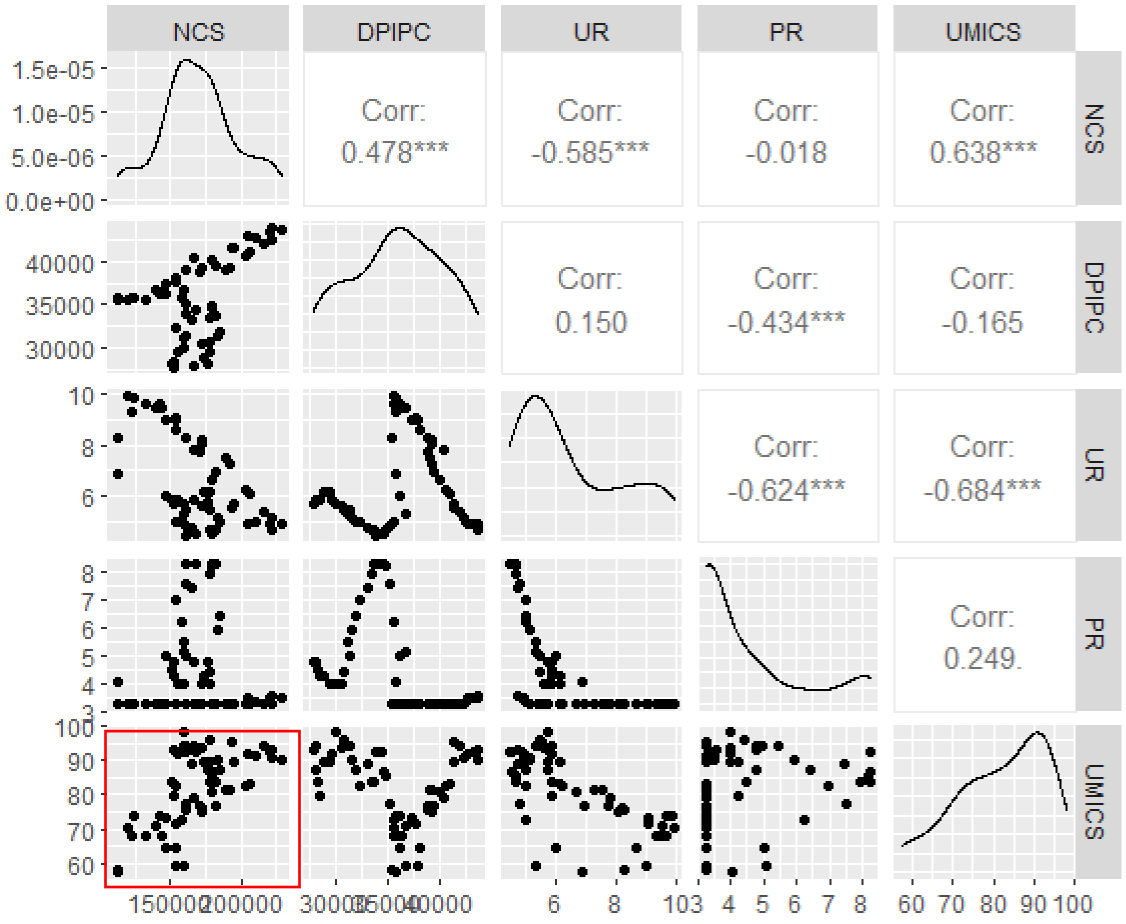



NCS and UMICS
Positive or negative relationship?

```
ncs %>%
  pivot_longer(-Quarter) %>%
  ggplot(aes(x= Quarter, y = value, color = name)) +
  geom_line() +
  facet_grid(name~., scales = "free_y")          # try without facet
```

# Scatterplots & correlation coefficients

```
ncs %>%
  GGally::ggpairs(columns = 2:6)
```




NCS and UMICS
Positive or negative relationship?

# Example: Regression equation

**Regression equation**

$$\hat{y} = b_0 + b_1 x_1 + b_2 x_2 + \cdots + b_k x_k$$

where

- $b_0$ is the **constant term**—the value of $y$ when the values of all the $x$ variables are set at 0;
- $b_1, \ldots, b_k$ are the **regression coefficients** (**slopes**)

Regression equation

NCS.Forecast = 25428.2554 + 3.2275*DPIPC – 7666.8516*UR –3219.4991*PR + 1122.4290*UMICS

```
fit <- ncs %>%
  model(tslm = TSLM(NCS ~ DPIPC + UR + PR + UMICS))

report(fit)
```

```
Coefficients:
              Estimate
(Intercept) 25428.2554
DPIPC           3.2275
UR          -7666.8516
PR          -3219.4991
UMICS        1122.4290
```

| | *Coefficients* |
|---|---|
| Intercept | 25428.25537 |
| DPIPC | 3.227461566 |
| UR | -7666.851578 |
| PR | -3219.499125 |
| UMICS | 1122.429026 |

Excel

Data Analysis → Regression

# Example: Regression equation Coefficient interpretation

Regression equation

NCS.Forecast = 25428.2554 + 3.2275*DPIPC – 7666.8516*UR −3219.4991*PR + 1122.4290*UMICS

Regression equation

$$\hat{y} = b_0 + b_1x_1 + b_2x_2 + \cdots + b_kx_k$$

where

- $b_0$ is the constant term—the value of $y$ when the values of all the $x$ variables are set at 0

- $b_1, \dots, b_k$ are the regression coefficients (slopes), indicating the change in the value of $y$ for each unit of change in the relevant $x$ value *ceteris paribus* (given that all other things being equal)
  - Holding other variables (DPIPC, UR, PR) constant, a one-unit increase in UMICS (the consumer sentiment index) is associated with an increase of 1122.43 units in NCS (number of cars sold) on average.

# Excel's Data Analysis: Regression

| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | Date | NCS | DPIPC | UR | PR | UMICS |
| 2 | 02-Feb | 152641 | 27840 | 5.7 | 4.75 | 93.13 |
| 3 | 02-May | 166036 | 28134 | 5.83 | 4.75 | 94.1 |
| 4 | 02-Aug | 175863 | 28168 | 5.73 | 4.75 | 87.27 |
| 5 | 02-Nov | 151219 | 28363 | 5.87 | 4.45 | 83.83 |
| 6 | 03-Feb | 152843 | 28584 | 5.87 | 4.25 | 79.97 |
| 7 | 03-May | 173360 | 28958 | 6.13 | 4.24 | 89.27 |
| 8 | 03-Aug | 178086 | 29539 | 6.13 | 4 | 89.3 |
| 9 | 03-Nov | 154916 | 29708 | 5.83 | 4 | 91.97 |
| 10 | 04-Feb | 159086 | 30094 | 5.7 | 4 | 98 |
| 11 | 04-May | 172174 | 30537 | 5.6 | 4 | 93.33 |
| 12 | 04-Aug | 178077 | 30800 | 5.43 | 4.42 | 95.6 |
| 13 | 04-Nov | 161216 | 31353 | 5.43 | 4.94 | 93.87 |
| 14 | 05-Feb | 160070 | 31145 | 5.3 | 5.44 | 94.07 |
| 15 | 05-May | 183594 | 31533 | 5.1 | 5.91 | 90.2 |
| 16 | 05-Aug | 184878 | 31961 | 4.97 | 6.43 | 87.5 |
| 17 | 05-Nov | 153405 | 32396 | 4.97 | 6.97 | 82.43 |
| 18 | 06-Feb | 164864 | 33217 | 4.73 | 7.43 | 88.93 |
| 19 | 06-May | 178363 | 33448 | 4.63 | 7.9 | 83.8 |
| 20 | 06-Aug | 181615 | 33696 | 4.63 | 8.25 | 84.03 |
| 21 | 06-Nov | 160780 | 33991 | 4.43 | 8.25 | 92.47 |
| 22 | 07-Feb | 167528 | 34457 | 4.5 | 8.25 | 92.2 |
| 23 | 07-May | 180006 | 34720 | 4.5 | 8.25 | 86.9 |

| | H | I | J | K | L |
|---|---|---|---|---|---|
| 1 | SUMMARY OUTPUT | | | | |
| 2 | | | | | |
| 3 | *Regression Statistics* | | | | |
| 4 | Multiple F | 0.910035 | | | |
| 5 | R Square | 0.828163 | | | |
| 6 | Adjusted I | 0.815666 | | | |
| 7 | Standard E | 11689.89 | | | |
| 8 | Observati | 60 | | | |
| 9 | | | | | |
| 10 | ANOVA | | | | |
| 11 | | *df* | *SS* | *MS* | *F* |
| 12 | Regressio | 4 | 3.62E+10 | 9.06E+09 | 66.26779 |
| 13 | Residual | 55 | 7.52E+09 | 1.37E+08 | |
| 14 | Total | 59 | 4.37E+10 | | |
| 15 | | | | | |
| 16 | | *Coefficients* | *tandard Err* | *t Stat* | *P-value* |
| 17 | Intercept | 25428.26 | 37465.77 | 0.678706 | 0.50017 |
| 18 | DPIPC | 3.227462 | 0.391571 | 8.242334 | 3.52E-11 |
| 19 | UR | -7666.85 | 1662.402 | -4.61191 | 2.42E-05 |
| 20 | PR | -3219.5 | 1412.531 | -2.27924 | 0.026556 |
| 21 | UMICS | 1122.429 | 210.6755 | 5.327763 | 1.91E-06 |

| | N | O | P | Q | R | S | T |
|---|---|---|---|---|---|---|---|
| 2 | Coefficients | 25428.26 | 3.227462 | -7666.85 | -3219.5 | 1122.429 | |
| 3 | Seasonal Naïve for 2017 | Ones | DPIPC | UR | PR | UMICS | NCS.Forecast |
| 4 | Q1 | 1 | 42807 | 4.93 | 3.5 | 91.57 | 217301.2 |
| 5 | Q2 | 1 | 43265 | 4.87 | 3.5 | 92.4 | 220171.0 |
| 6 | Q3 | 1 | 43651 | 4.9 | 3.5 | 90.33 | 218863.4 |

```
Series: NCS
Model: TSLM

Residuals:
     Min       1Q   Median       3Q      Max
-27502.7  -8272.9    488.2   7831.5  20792.9

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) 25428.2554 37465.7657   0.679   0.5002
DPIPC           3.2275     0.3916   8.242 3.52e-11 ***
UR          -7666.8516  1662.4019  -4.612 2.42e-05 ***
PR          -3219.4991  1412.5314  -2.279   0.0266 *
UMICS        1122.4290   210.6755   5.328 1.91e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11690 on 55 degrees of freedom
Multiple R-squared: 0.8282,     Adjusted R-squared: 0.8157
F-statistic: 66.27 on 4 and 55 DF, p-value: < 2.22e-16
```

# Forecasting with regression

Scenario-based forecast

ex-ante vs ex-post forecasts

# TS Forecasting with Regression

| Date | NCS | DPIPC | UR | PR | UMICS |
|---|---|---|---|---|---|
| Feb-02 | 152641 | 27840 | 5.7 | 4.75 | 93.13 |
| May-02 | 166036 | 28134 | 5.83 | 4.75 | 94.1 |
| Aug-02 | 175863 | 28168 | 5.73 | 4.75 | 87.27 |
| Nov-02 | 151219 | 28363 | 5.87 | 4.45 | 83.83 |
| ... | | | | | |
| Feb-16 | 204681 | 42807 | 4.93 | 3.5 | 91.57 |
| May-16 | 221073 | 43265 | 4.87 | 3.5 | 92.4 |
| Aug-16 | 229123 | 43651 | 4.9 | 3.5 | 90.33 |
| Nov-16 | 221392 | 43759 | 4.7 | 3.55 | 93.07 |

| | *Coefficients* |
|---|---|
| Intercept | 25428.25537 |
| DPIPC | 3.227461566 |
| UR | -7666.851578 |
| PR | -3219.499125 |
| UMICS | 1122.429026 |

- Regression equation

NCS.Forecast = 25428.2554 + 3.2275*DPIPC – 7666.8516*UR –3219.4991*PR + 1122.4290*UMICS

- Suppose that in 2017 Q1, the forecasts of independent variables (i.e., DPIPC, UR, PR, UMICS) come from seasonal naïve.

DPIPC = 42807, UR = 4.93, PR = 3.50, UMICS = 91.57.

  - Different TS forecasting methods can be used for different series.

- Then, the NCS forecast for 2017Q1 is

NCS.Forecast = 25428.2554 + 3.2275(42807) – 7666.8516(4.93) –3219.4991(3.50) + 1122.4290*(91.57)

= 217301

Forecast 2017 Q2, Q3, Q4

# Forecasting with Regression Forecasts of predictors

- We require the forecasts of the predictors.
- To obtain these forecasts,
  - We can use one of the simple methods or more sophisticated pure time series approaches.
  - Alternatively, forecasts from other sources may be available and can be used.
    - Forecasts from government agency
      - GDP
      - Weather
    - Forecasts from sales department in the Sales and Operations Planning (S&OP) meeting.

# R: TS cross validation to select forecast for predictors*

```r
# "unpivot": long format
ncs_long <- ncs %>%
  select(-NCS) %>%
  pivot_longer(!Quarter, names_to="variable", values_to = "value")
ncs_long

trCV <- ncs_long %>%
  slice(1:(n()-4*4)) %>%
  stretch_tsibble(.init=6, .step=1) %>%
  relocate(.id, Quarter)

trCV %>%
  model(
    Mean = MEAN(value),
    Naive = NAIVE(value),
    Drift = NAIVE(value ~ drift()),
    Snaive = SNAIVE(value)
  ) %>%
  forecast(h=4) %>%
  accuracy(ncs_long) %>%
  arrange(variable, RMSE)
```

`!Quarter` All data except column `Quarter`

```
# A tsibble: 240 x 3 [1Q]
# Key:       variable [4]
   Quarter variable   value
     <qtr> <chr>      <dbl>
 1 2002 Q1 DPIPC    27840
 2 2002 Q1 UR           5.7
 3 2002 Q1 PR           4.75
 4 2002 Q1 UMICS       93.1
 5 2002 Q2 DPIPC    28134
 6 2002 Q2 UR           5.83
 7 2002 Q2 PR           4.75
 8 2002 Q2 UMICS       94.1
 9 2002 Q3 DPIPC    28168
10 2002 Q3 UR           5.73
# ... with 230 more rows
```

| | .model | variable | .type | ME | RMSE | MAE | MPE | MAPE | MASE | RMSSE | ACF1 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | <chr> | <chr> | <chr> | <dbl> | <dbl> | <dbl> | <dbl> | <dbl> | <dbl> | <dbl> | <dbl> |
| 1 | Drift | DPIPC | Test | -28.7 | 634. | 430. | -0.0849 | 1.17 | 0.360 | 0.484 | 0.664 |
| 2 | Naive | DPIPC | Test | 677. | 955. | 814. | 1.85 | 2.22 | 0.680 | 0.729 | 0.485 |
| 3 | Snaive | DPIPC | Test | 1088. | 1324. | 1213. | 3.00 | 3.33 | 1.01 | 1.01 | 0.605 |
| 4 | Mean | DPIPC | Test | 4649. | 4950. | 4649. | 12.3 | 12.3 | 3.88 | 3.78 | 0.915 |
| 5 | Naive | PR | Test | -0.0290 | 0.893 | 0.471 | -2.19 | 9.27 | 0.616 | 0.708 | 0.792 |
| 6 | Drift | PR | Test | -0.0409 | 0.977 | 0.560 | -1.34 | 11.6 | 0.733 | 0.775 | 0.779 |
| 7 | Snaive | PR | Test | -0.0622 | 1.27 | 0.755 | -4.53 | 15.1 | 0.988 | 1.00 | 0.915 |
| 8 | Mean | PR | Test | -0.543 | 1.92 | 1.76 | -26.4 | 42.8 | 2.30 | 1.52 | 0.967 |
| 9 | Naive | UMICS | Test | 0.0504 | 7.61 | 5.70 | -0.486 | 7.57 | 0.727 | 0.757 | 0.678 |
| 10 | Drift | UMICS | Test | 1.11 | 8.10 | 6.25 | 0.978 | 8.21 | 0.797 | 0.807 | 0.682 |
| 11 | Snaive | UMICS | Test | 0.319 | 9.85 | 7.56 | -0.528 | 10.0 | 0.964 | 0.981 | 0.669 |
| 12 | Mean | UMICS | Test | -4.24 | 12.0 | 9.39 | -7.34 | 12.9 | 1.20 | 1.20 | 0.876 |
| 13 | Naive | UR | Test | -0.0561 | 0.873 | 0.579 | -1.77 | 8.43 | 0.658 | 0.712 | 0.796 |
| 14 | Drift | UR | Test | -0.113 | 0.926 | 0.644 | -2.18 | 9.26 | 0.732 | 0.756 | 0.786 |
| 15 | Snaive | UR | Test | -0.0780 | 1.24 | 0.904 | -2.92 | 13.0 | 1.03 | 1.01 | 0.909 |
| 16 | Mean | UR | Test | 0.575 | 1.93 | 1.50 | 2.44 | 21.0 | 1.71 | 1.58 | 0.966 |

Based on smallest RMSE

| Predictor | FC |
|---|---|
| DPIPC | Drift |
| PR | Naïve |
| UMIC | Naïve |
| UR | Naïve |

Also try ARIMA and ETS

# “What-If” Scenarios

- Suppose the sales of a company depend on the price it charges and the amount it spends on different media:

Sales = 120 – 0.4(Avg Price) + 5(Media Spending)

where sales and media spending are given in millions of dollars

- Current plan: Avg price = 10 USD/unit, Media spending = 2 million.
  - Sales.Forecast = 120 – 0.4(10) + 5(2) = 126 million
- But, management wants to hit the sales target of 130 million.
- If we keep avg price and change media spending, how much must be spent on the media to hit the target?

130 = 120 – 0.4(10) + 5($x_2$) → $x_2$ =2.8 million

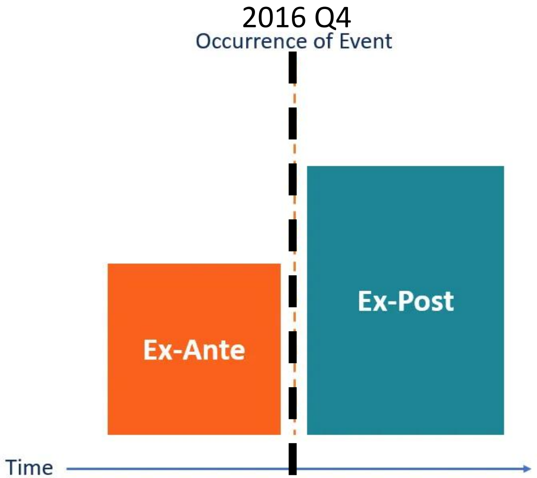
# Ex-ante *vs* Ex-post forecasts

## Ex-ante uses forecasted predictors, Ex-post uses actual predictors

### Ex-ante forecast

Ex-ante forecasts are those that are made using only the information that is available.

- Based on forecasts rather than actual results.




### Ex-post forecast

Ex-post forecasts are those that are made using later information on the predictors.

- Based on actual results rather than forecasts

| | Intercept | DPIPC | UR | PR | UMICS | NCS.FC |
|---|---|---|---|---|---|---|
| | 25428.26 | 3.227462 | -7666.85 | -3219.5 | 1122.429 | |
| | | | | | | |
| 2017Q1 | | 42807 | 4.93 | 3.5 | 91.57 | 217,301 |
| 2017Q2 | | [illegible] | [illegible] | [illegible] | [illegible] | 220,171 |
| 2017Q3 | | [illegible] | [illegible] | [illegible] | [illegible] | 218,863 |
| 2017Q4 | | 43759 | 4.7 | 3.55 | 93.07 | 223,660 |

Forecasts using data up to 2016

NCS.FC = 25428.26 + 3.23*DPIPC.FC – 7666.85*UR.FC –3219.5*PR.FC + 1122.43*UMICS.FC

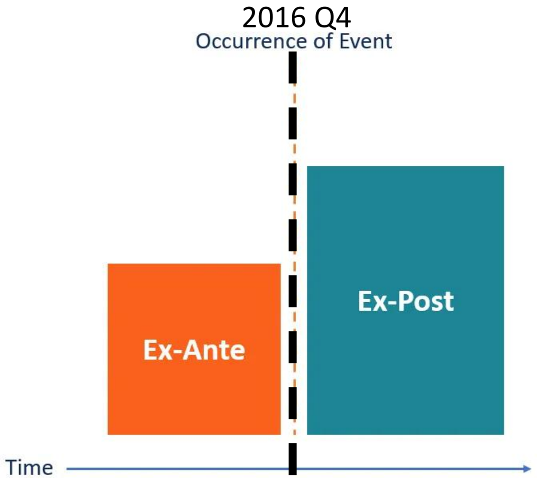
| | Intercept | DPIPC | UR | PR | UMICS | NCS.FS |
|---|---|---|---|---|---|---|
| | 25428.26 | 3.227462 | -7666.85 | -3219.5 | 1122.429 | |
| | | | | | | |
| 2017Q1 | | 42807 | 4.97 | 3.4 | 92.73 | 218,618 |
| 2017Q2 | | [illegible] | [illegible] | [illegible] | [illegible] | 218,314 |
| 2017Q3 | | [illegible] | [illegible] | [illegible] | [illegible] | 223,220 |
| 2017Q4 | | 43759 | 4.6 | 3.4 | 5.1 | 126,169 |

Actual values in 2017

NCS.FC = 25428.26 + 3.23*DPIPC.Actual – 7666.85*UR.Actual –3219.0*PR.Actual + 1122.43*UMICS.Actual

# Ex-ante vs Ex-post forecasts

- A comparative evaluation of ex-ante forecasts and ex-post forecasts can help to separate out the sources of forecast uncertainty.
- This will show whether forecast errors have arisen
  a) due to poor forecasts of the predictor or
  b) [*model misspecification*] due to a poor forecasting model (regression model, here).




## Ex-ante forecast

| | Intercept | DPIPC | UR | PR | UMICS | NCS.FC |
|---|---|---|---|---|---|---|
| | 25428.26 | 3.227462 | -7666.85 | -3219.5 | 1122.429 | |
| | | | | | | |
| 2017Q1 | | 42807 | 4.93 | 3.5 | 91.57 | 217,301 |
| 2017Q2 | | [illegible] | [illegible] | [illegible] | [illegible] | 220,171 |
| 2017Q3 | | [illegible] | [illegible] | [illegible] | [illegible] | 218,863 |
| 2017Q4 | | 43759 | 4.7 | 3.55 | 93.07 | 223,660 |

Forecasts using data up to 2016

NCS.FC = 25428.26 + 3.23*DPIPC.FC – 7666.85*UR.FC –3219.5*PR.FC + 1122.43*UMICS.FC

## Ex-post forecast

| | Intercept | DPIPC | UR | PR | UMICS | NCS.FS |
|---|---|---|---|---|---|---|
| | 25428.26 | 3.227462 | -7666.85 | -3219.5 | 1122.429 | |
| | | | | | | |
| 2017Q1 | | 42807 | 4.97 | 3.4 | 92.73 | 218,618 |
| 2017Q2 | | [illegible] | [illegible] | [illegible] | [illegible] | 218,314 |
| 2017Q3 | | [illegible] | [illegible] | [illegible] | [illegible] | 223,220 |
| 2017Q4 | | 43759 | 4.6 | 3.4 | 5.1 | 126,169 |

Actual values in 2017

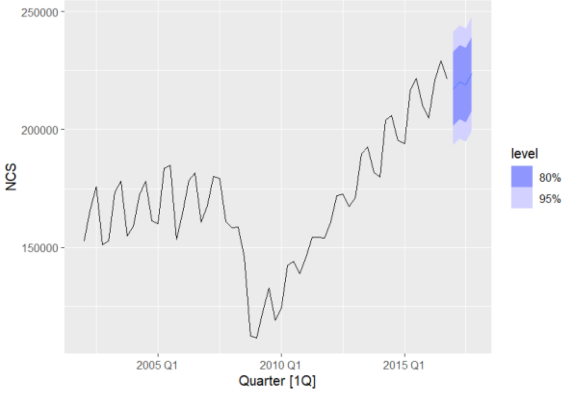
NCS.FC = 25428.26 + 3.23*DPIPC.Actual – 7666.85*UR.Actual –3219.0*PR.Actual + 1122.43*UMICS.Actual

Example
I) due to poor forecasts of the predictor: MAPE ex-ante forecast 60%, while MAPE ex-post forecast 8%
II) due to a poor forecasting model: MAPE ex-ante forecast 60%, while MAPE ex-post forecast 58%

# Ex-ante forecast: Prediction interval*

- The prediction intervals fore ex-ante forecasts do not include the uncertainty associated with the future values of the predictor variables.
- They assume that the values of the predictors are known in advance.

```
ncs_tr <- ncs %>%
  filter(year(Quarter) <= 2015)

ncs_test <- ncs %>%
  filter(year(Quarter) >= 2016)

fit <- ncs_tr %>%
  model(tslm = TSLM(NCS ~ DPIPC + UR + PR + UMICS))

fc_predictor <- tsibble(
    Quarter = yearquarter("2016 Q1") + 0:3,
    DPIPC = rep(42621, 4),
    PR = rep( 3.29, 4),
    UMICS = rep(91.3, 4),
    UR = rep(5, 4),
    index = Quarter
  )
fc_exante <- forecast(fit, new_data = fc_predictor)

fc_exante %>%
  hilo(level=99)
```




```
  .model Quarter                 NCS   .mean DPIPC    PR UMICS    UR                `99%`
  <chr>    <qtr>              <dist>   <dbl> <dbl> <dbl> <dbl> <dbl>               <hilo>
1 tslm   2016 Q1 N(217037, 1.6e+08) 217037. 42621  3.29  91.3     5 [184181.4, 249893]99
2 tslm   2016 Q2 N(217037, 1.6e+08) 217037. 42621  3.29  91.3     5 [184181.4, 249893]99
3 tslm   2016 Q3 N(217037, 1.6e+08) 217037. 42621  3.29  91.3     5 [184181.4, 249893]99
4 tslm   2016 Q4 N(217037, 1.6e+08) 217037. 42621  3.29  91.3     5 [184181.4, 249893]99
```

# Regression model: Assumptions

# Multiple Linear Regression: Assumptions

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_k x_{ki} + \epsilon_i$$

- The relationship between $y$ and $x_k$ is linear and given above.
- The $x$'s are nonstochastic variables. In addition, *no exact linear relationship* exists between two or more independent variables.
- Errors $\epsilon_i \sim NID(0, \sigma_e^2)$
  - The error has zero expected value for all observations
  - **Homoscedasticity**: Error term has constant variance for all observations.
  - **White noise**: Errors corresponding to different observations are independent and therefore uncorrelated. No serial correlation.
  - The error term is normally distributed.

# Diagnostic Test

1. Do signs of coefficients make sense?
2. Is the coefficient statistically significant?
3. Explanatory power: **Adjusted R-Square**
4. Residual analysis
5. Multicollinearity

```
Series: NCS
Model: TSLM

Residuals:
     Min       1Q   Median       3Q      Max
-27502.7  -8272.9    488.2   7831.5  20792.9

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) 25428.2554 37465.7657   0.679   0.5002
DPIPC           3.2275     0.3916   8.242 3.52e-11 ***
UR          -7666.8516  1662.4019  -4.612 2.42e-05 ***
PR          -3219.4991  1412.5314  -2.279   0.0266 *
UMICS        1122.4290   210.6755   5.328 1.91e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 11690 on 55 degrees of freedom
Multiple R-squared: 0.8282,     Adjusted R-squared: 0.8157
F-statistic: 66.27 on 4 and 55 DF, p-value: < 2.22e-16
```

| *Regression Statistics* | |
|---|---|
| Multiple R | 0.910034731 |
| R Square | 0.828163212 |
| Adjusted R | 0.815665991 |
| Standard E | 11689.89219 |
| Observatic | 60 |

ANOVA

| | *df* | *SS* | *MS* | *F* | *ignificance F* |
|---|---|---|---|---|---|
| Regressior | 4 | 3.62E+10 | 9.06E+09 | 66.26779 | 2.2E-20 |
| Residual | 55 | 7.52E+09 | 1.37E+08 | | |
| Total | 59 | 4.37E+10 | | | |

| | *Coefficients* | *andard Err* | *t Stat* | *P-value* | *Lower 95%* | *Upper 95%* | *ower 95.0%* | *pper 95.0%* |
|---|---|---|---|---|---|---|---|---|
| Intercept | 25428.25537 | 37465.77 | 0.678706 | 0.50017 | -49654.8 | 100511.3 | -49654.8 | 100511.3 |
| DPIPC | 3.227461566 | 0.391571 | 8.242334 | 3.52E-11 | 2.442735 | 4.012188 | 2.442735 | 4.012188 |
| UR | -7666.851578 | 1662.402 | -4.61191 | 2.42E-05 | -10998.4 | -4335.32 | -10998.4 | -4335.32 |
| PR | -3219.499125 | 1412.531 | -2.27924 | 0.026556 | -6050.28 | -388.723 | -6050.28 | -388.723 |
| UMICS | 1122.429026 | 210.6755 | 5.327763 | 1.91E-06 | 700.226 | 1544.632 | 700.226 | 1544.632 |

# 1. Logic: Check the signs of the coefficients

Predicted Sale = 25304.35 + 3.227*DPIPC – 7659.605*UR -3216.77*PR + 1123.363*UMICS

Dependent variable

- NCS = New car sales

Independent variables

- DPIPC = Disposable income per capita
- UMICS = University of Michigan Index of Consumer Sentiment
- UR = Unemployment rate
- PR = Prime interest rate

These coefficients measure the marginal effects, i.e., the effect of each predictor given all other things being equal (*ceteris paribus*).

# 2. Check the significance of the coefficients Example: NCS

- Null hypothesis $H_0: \beta_i = 0$
- Alternative hypothesis $H_1: \beta_i \neq 0$

- The ***t* statistic** is the ratio of an estimated coefficient to its standard error.

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept) 25304.3496 37454.5300   0.676   0.5021    
DPIPC           3.2273     0.3916   8.240 3.54e-11 ***
UR          -7659.6055  1661.5855  -4.610 2.44e-05 ***
PR          -3216.7766  1412.4494  -2.277   0.0267 *  
UMICS        1123.3628   210.5820   5.335 1.86e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- For DPIPC, $\hat{\beta}_1 = 3.2273$, and its standard error is 0.3916, the $t$ value = $3.2273/0.3916$ = 8.24

# 2. Check the significance of the coefficients Example: NCS

- Null hypothesis $H_0: \beta_i = 0$
- Alternative hypothesis $H_1: \beta_i \neq 0$

- The ***t* statistic** follows the t distribution with the degrees of freedom, $T_{\mathrm{df}}$

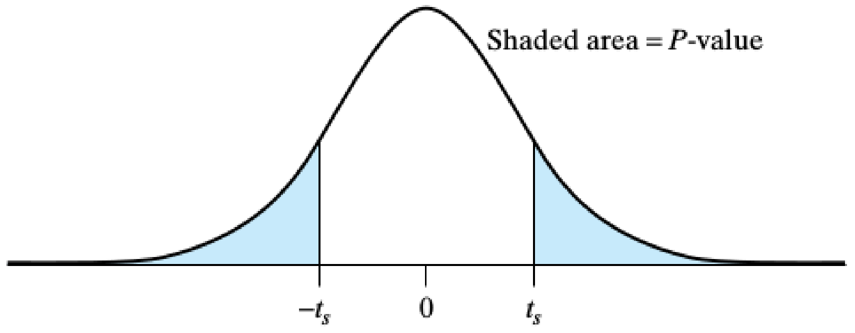
df = $n - (k+1) = 60 - (4+1) = 55.$

- For PR, $t = -2.277$, the p-value is $\Pr(|T_{55}| > 2.277) = 2(0.01335) = 0.0267$

```
> 1-pt(2.277, 55)  # right tail
[1] 0.01334926
> 2*(1-pt(2.277, 55))  # two tail
[1] 0.02669851
```

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 25304.3496 37454.5300   0.676   0.5021
DPIPC           3.2273     0.3916   8.240 3.54e-11 ***
UR          -7659.6055  1661.5855  -4.610 2.44e-05 ***
PR          -3216.7766  1412.4494  -2.277   0.0267 *
UMICS        1123.3628   210.5820   5.335 1.86e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Three stars (or asterisks) represent a highly significant p-value.




$H_0: \beta_4 = 0$ vs $H_1: \beta_4 \neq 0$

Let significance level $\alpha = 0.05$. Since p-value = $1.86 \times 10^{-6} < \alpha$, there is sufficient evidence to reject the null hypothesis $H_0$ in favor of $H_1$.

Remove PR: Re-do the regression.

# 3. Coefficient of determination $R^2$

| TABLE 6 | ANOVA Table for Multiple Regression | | | |
|---|---|---|---|---|
| *Source* | *Sum of Squares* | *df* | *Mean Square* | *F Ratio* |
| Regression | $SSR$ | $k$ | $MSR = SSR/k$ | $F = \frac{MSR}{MSE}$ |
| Error | $SSE$ | $n-k-1$ | $MSE = SSE/(n-k-1)$ | |
| Total | $SST$ | $n-1$ | | |

Error sum of squares SSE = $\sum(y_i - \hat{y}_i)^2$
Total sum of squares SST = $\sum(y_i - \bar{y})^2$

$$R^2 = \frac{\text{Explained variation in Y}}{\text{Total variation in Y}} = \frac{SSR}{SST}$$

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2} = 1 - \frac{\text{SSE}}{\text{SST}}$$

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 25304.3496 37454.5300   0.676   0.5021
DPIPC           3.2273     0.3916   8.240 3.54e-11 ***
UR          -7659.6055  1661.5855  -4.610 2.44e-05 ***
PR          -3216.7766  1412.4494  -2.277   0.0267 *
UMICS        1123.3628   210.5820   5.335 1.86e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 11690 on 55 degrees of freedom
Multiple R-squared:  0.8281,    Adjusted R-squared:  0.8156
F-statistic: 66.26 on 4 and 55 DF,  p-value: < 2.2e-16
```

| *Regression Statistics* | |
|---|---|
| Multiple R | 0.910034731 |
| R Square | 0.828163212 |
| Adjusted R Square | 0.815665991 |
| Standard Error | 11689.89219 |
| Observations | 60 |

The model explains 82.81% of the variation in NCS.

# Adjusted $R^2$: $\bar{R}^2$

- Adding any variable increases the value of $R^2$, even if that variable is irrelevant.
- For these reasons, forecasters should not use $R^2$ to determine whether a model will give good predictions, as it will lead to ***overfitting***.
- An alternative which is designed to overcome these problems is the **adjusted $\boldsymbol{R^2}$.**

$$\bar{R}^2 = 1 - (1 - R^2)\frac{n-1}{n-k-1}$$

where $n$ is the number of observations, and $k$ is the number of predictors.

- Adjusted $R^2$ no longer increases with each added predictor.
- Using this measure, the best model will be the one with the largest value of adjusted $R^2$.

# Adjusted $R^2$
# Underfitting vs overfitting

```
> fitUnder <- ncs %>%
+   model(tslm = TSLM(NCS ~ DPIPC))
>
> report(fitUnder)
Series: NCS
Model: TSLM

Residuals:
    Min      1Q Median      3Q     Max
-56557 -15584   3961  18723  37605

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 6.640e+04  2.505e+04   2.651 0.010329 *
DPIPC       2.866e+00  6.917e-01   4.144 0.000112 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 24120 on 58 degrees of freedom
Multiple R-squared: 0.2284,	Adjusted R-squared: 0.2151
F-statistic: 17.17 on 1 and 58 DF, p-value: 0.00011246
```

```
> fit <- ncs %>%
+   model(tslm = TSLM(NCS ~ DPIPC + UR + PR + UMICS))
>
> report(fit)
Series: NCS
Model: TSLM

Residuals:
     Min       1Q   Median       3Q      Max
-27502.7  -8272.9    488.2   7831.5  20792.9

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) 25428.2554 37465.7657   0.679   0.5002
DPIPC           3.2275     0.3916   8.242 3.52e-11 ***
UR          -7666.8516  1662.4019  -4.612 2.42e-05 ***
PR          -3219.4991  1412.5314  -2.279   0.0266 *
UMICS        1122.4290   210.6755   5.328 1.91e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 11690 on 55 degrees of freedom
Multiple R-squared: 0.8282,	Adjusted R-squared: 0.8157
F-statistic: 66.27 on 4 and 55 DF, p-value: < 2.22e-16
```

```
> ncs <- ncs %>%
+   mutate(URPR = UR*PR)
> fitOver <- ncs %>%
+   model(tslm = TSLM(NCS ~ DPIPC + UR + PR + UMICS + URPR))
> report(fitOver)
Series: NCS
Model: TSLM

Residuals:
   Min     1Q Median     3Q    Max
-26857  -7770    506   7984  20958

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) 22495.4038 38639.1676   0.582 0.562861
DPIPC           3.0201     0.6995   4.318 6.79e-05 ***
UR          -3562.6781 11552.0453  -0.308 0.758963
PR           3019.8538 17434.4371   0.173 0.863132
UMICS        1064.7210   266.3214   3.998 0.000195 ***
URPR        -1451.6013  4042.6164  -0.359 0.720939
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 11780 on 54 degrees of freedom
Multiple R-squared: 0.8286,	Adjusted R-squared: 0.8127
F-statistic:  52.2 on 5 and 54 DF, p-value: < 2.22e-16
```

Adding any variable increases the value of $R^2$, even if that variable is irrelevant.

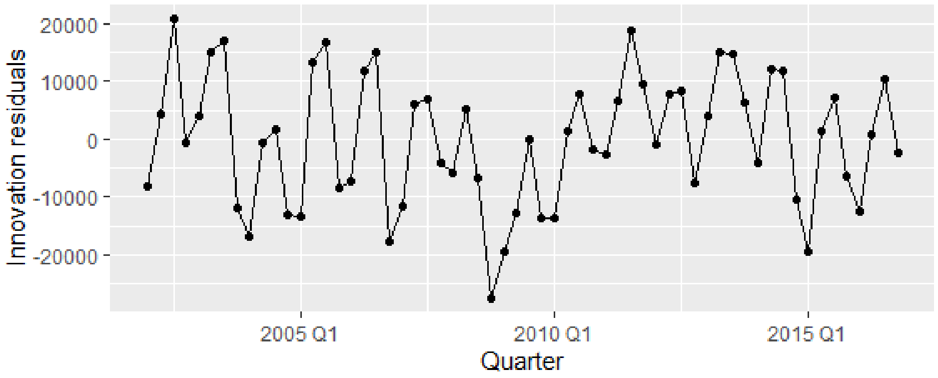
TRUE/FALSE: “$R^2$ tells you the % of variation explained by only the variables that *actually affect* the dependent variable”

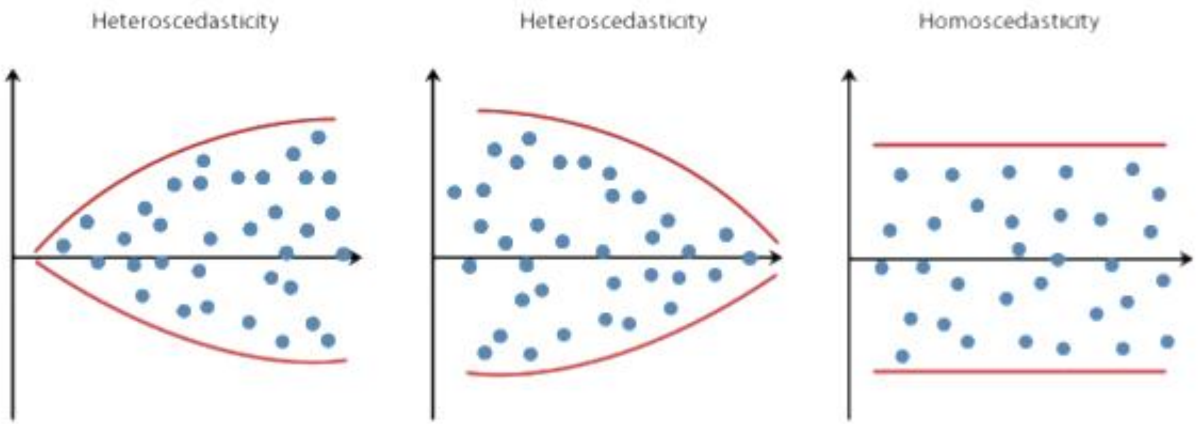
For these reasons, forecasters should not use $R^2$ to determine whether a model will give good predictions, as it will lead to ***overfitting***.

Adjusted $R^2$ no longer increases with each added predictor.
Using this measure, the best model will be the one with the largest value of adjusted $R^2$.

# 4. Check assumptions of errors

Homoscedasticity: No changing variation over time.

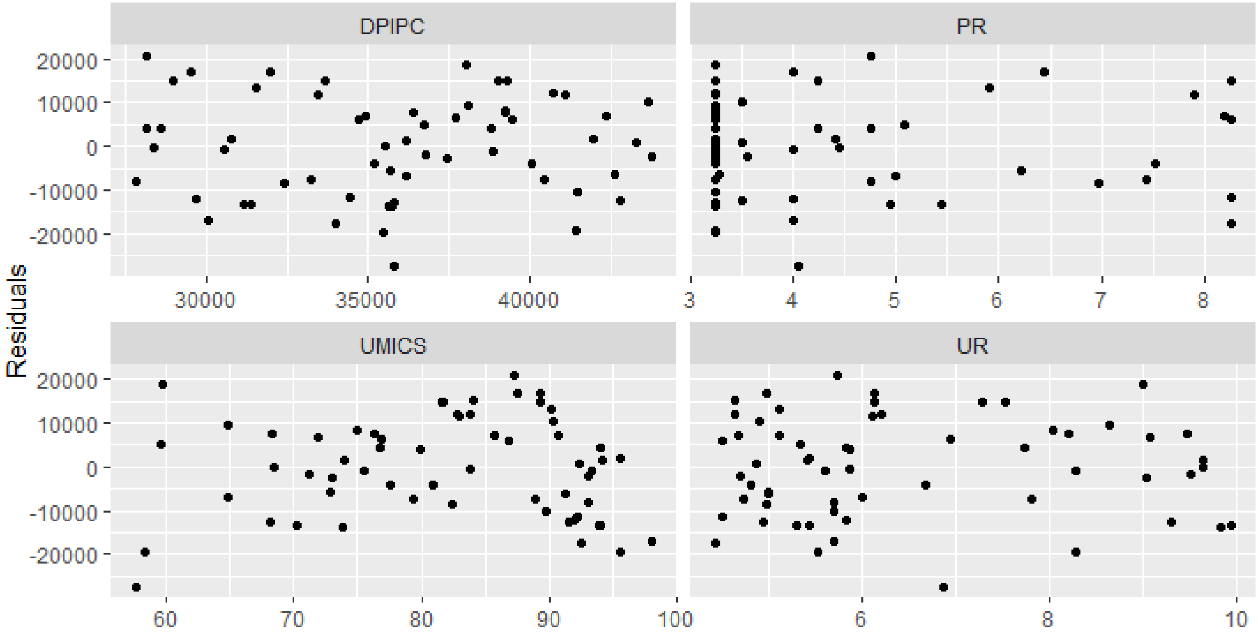


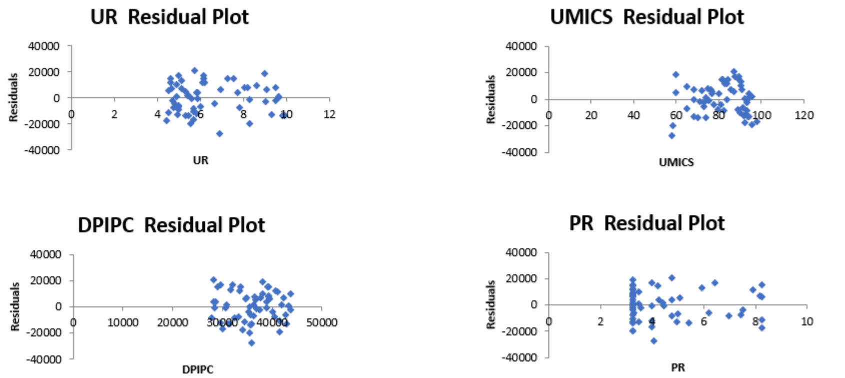
# Residual plots against predictors

```
ncs %>%
  left_join(residuals(fit), by = "Quarter") %>%
  pivot_longer(DPIPC:UMICS,
               names_to = "regressor", values_to = "x") %>%
  ggplot() +
  geom_point(aes(x=x, y=.resid)) +
  facet_wrap(. ~regressor, scales = "free_x") +
  labs(y="Residuals", x="")
```

- Examine scatterplots of residuals against each of the predictor variables
- If these scatter plots show a pattern, then the relationship may be nonlinear and the model will need to be modified.
- It is also necessary to plot the residuals against any predictors that are not in the model.
- If any of these show a pattern, then the corresponding predictor may need to be added.



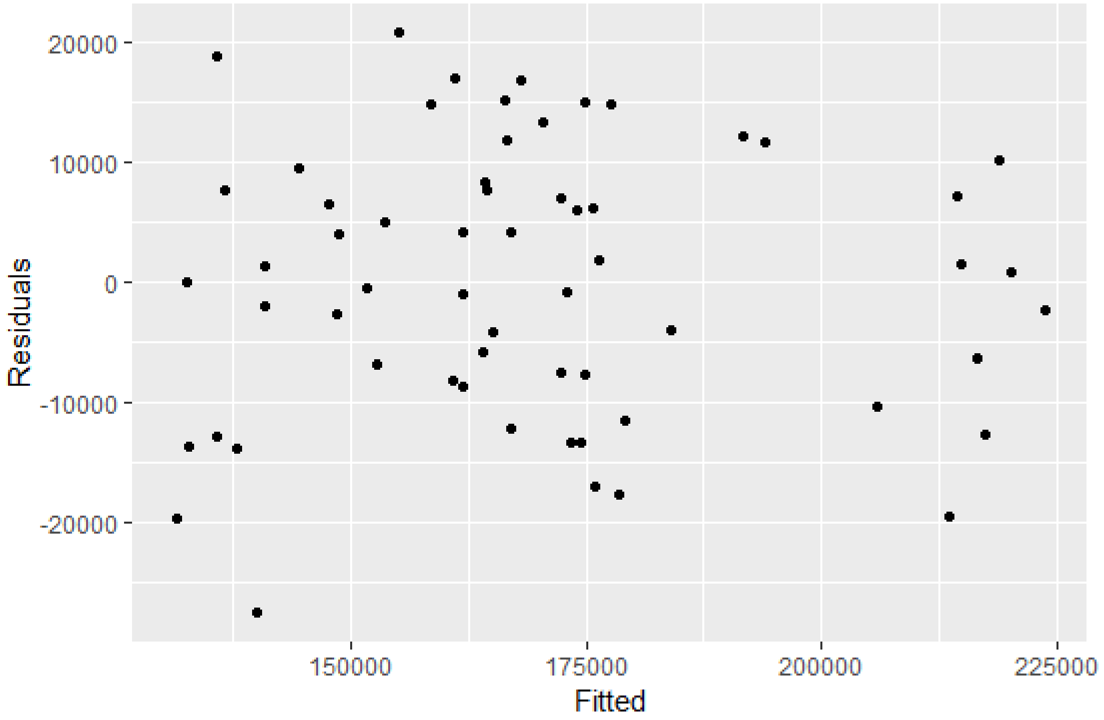

R




Excel

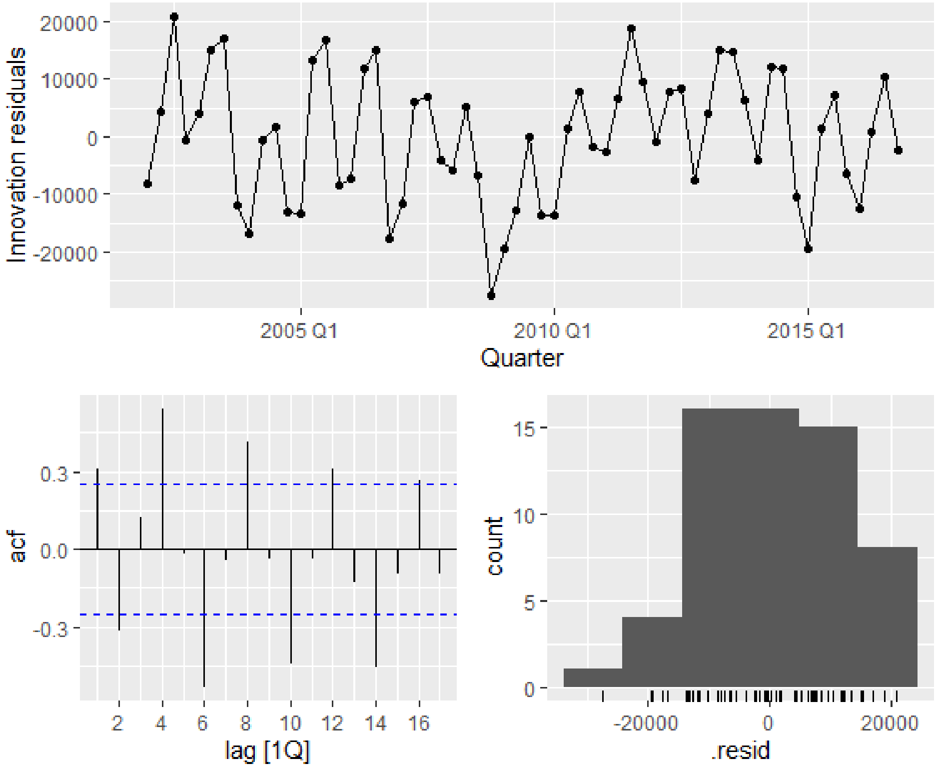
# Residual plots against fitted values

```
augment(fit) %>%
  ggplot() +
  geom_point(aes(x=.fitted, y = .resid)) +
  labs(x="Fitted", y="Residuals")
```




The random scatter suggests the errors are homoskedastic.

# 4. Check assumptions of errors: Serial correlation




```
fit %>% gg_tsresiduals()

augment(fit) %>%
  features(.innov, ljung_box, lag=10)
```

|   | .model | lb_stat | lb_pvalue |
|---|---|---|---|
|   | <chr> | <dbl> | <dbl> |
| 1 | tslm | 79.0 | 7.92e-13 |

Let $\rho_j = corr(\epsilon_t, \epsilon_{t-j})$ be the autocorrelation at lag $j$

$H_0: \rho_1 = \rho_2 = \cdots = \rho_{10} = 0$

At significance level $\alpha = 0.05$, there is sufficient evidence to reject $H_0$.

Serial correlation in errors: Error in one period is correlated with the errors in other periods.

# Serial Correlation

- When fitting a regression model to time series data, it is common to find autocorrelation in the residuals.
- In this case, the estimated model violates the assumption of no autocorrelation in the errors (i.e., errors are independent over time).
  - Our forecasts may be inefficient — there is some information left over which should be accounted for in the model in order to obtain better forecasts.
  - The most common reason for serial correlation is that an important explanatory variable has been *omitted*.
- The forecasts from a model with autocorrelated errors are still unbiased, and so are not "wrong", but they will usually have larger prediction intervals than they need to.

# 5. Multicollinearity

- In multiple-regression analysis, one of the assumptions that are made is that the independent variables are not highly linearly correlated with each other or with linear combinations of other independent variables.
- If this assumption is violated, a problem known as ***multicollinearity*** results.

| | DPIPC | GDP | IR |
|---|---|---|---|
| DPIPC | 1 | | |
| GDP | *0.99* | 1 | |
| IR | −0.65 | −0.67 | 1 |

- If we look at these correlations between these variables, we can see a problem.
- Clearly, there is a very strong linear association between GDP and DPIPC.
- In this case, both of these variables are measuring essentially the same thing.

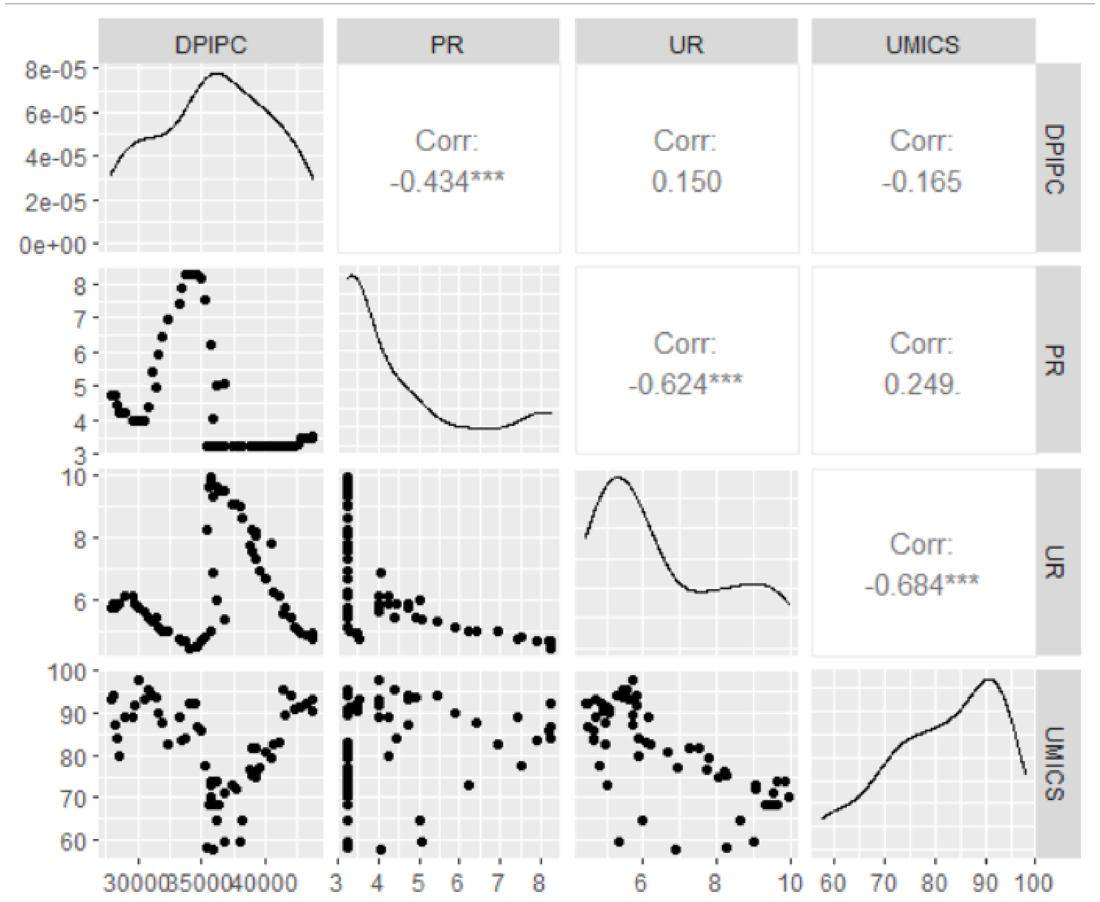
# Example
# 5. Multicollinearity

- Note that no pair of independent variables has a correlation that is close to 1.0.

- The correlations of 1.00 along the main diagonal are to be expected.
  - They simply tell us that each variable is perfectly correlated with itself.

```
ncs %>%
  GGally::ggpairs(columns = 3:6)
```

Excel: Data Analysis -> Correlation



# Multiple Regressions: Things to Keep in Mind

- Independent variables should not only be statistically valid but also theoretically sound.
  - Correlation is not causation
- Monitor regularly the performance of a model. It may be the time to re-specify it.
- Statistics (e.g., adjusted $R^2$, F and $t$) should be used as guide, *not* bible.
  - At times, you may find that even a model with one or two unsatisfactory statistics gives forecasts that are acceptable.
- **Principle of Parsimony**: If two models yield more or less the same result, the one with fewer variables should be preferred.

"Essentially, all models are wrong, but some are useful"

Your model *has to be* wrong…

… but that's o.k. if it's illuminating!

George E.P. Box

# Model selections

Useful predictors

Selecting predictors

# Model Selection

- **Best subset selection** or "all possible subsets" regression
  - Recall from slide 01, suppose that there are $k$ predictors, then the number of possible models is $2^k - 1$
- **Stepwise regression**
  - Backwards stepwise regression
  - Forward stepwise regression
  - Hybrid procedure
- Stepwise regression is a greedy heuristic algorithm
  - Not guaranteed to lead to the best possible model.

## Model Evaluation Example: Baseball

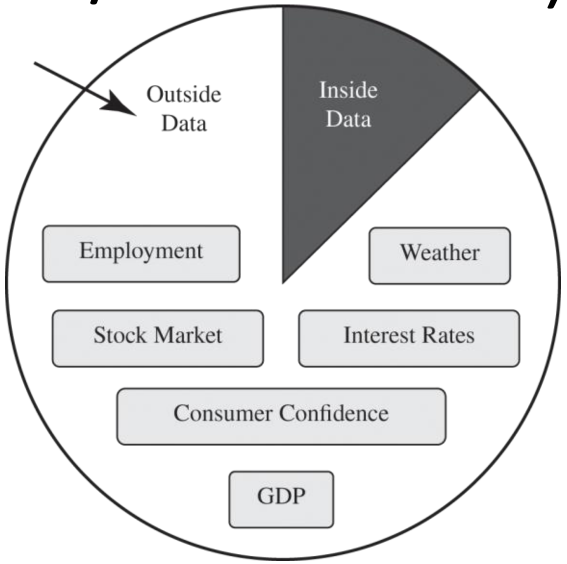
- Dependent variable: `Salary_K`
- All possible models: How many?

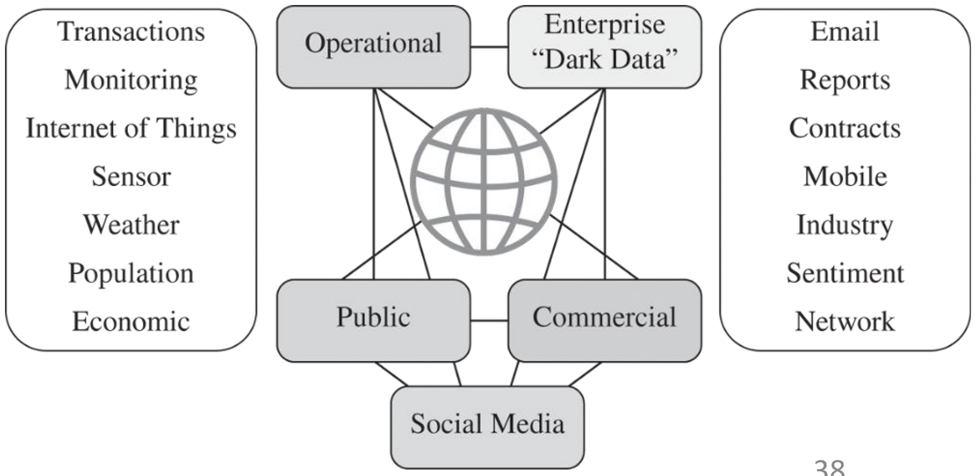
| | Years_in_Majors | Career_ERA | Innings_Pitched | Career_Wins | Career_Losses |
|---|---|---|---|---|---|
| 1 | 1 | 0 | 0 | 0 | 0 |
| 2 | 1 | 0 | 1 | 0 | 0 |
| 3 | 1 | 1 | 1 | 1 | 1 |
| | | | … | | |
| | | | | | |
| | | | | | |
| | | | | | |
| | | | | | |
| | | | | | |

# Demand Forecasting: Independent Variables (Predictors/Features)

- Internal/Endogeneous
  - Price of product
  - Media spending
  - Number of new products to be introduced
  - Number of products to be discontinued
  - Number of stores
  - FSI (Free Standing Insert) coupons distributed
- External/Exogeneous
  - Competitive price
  - New competitive product to be launched
  - Gross Domestic Product (GDP)
  - Consumer price index (CPI)
  - Unemployment

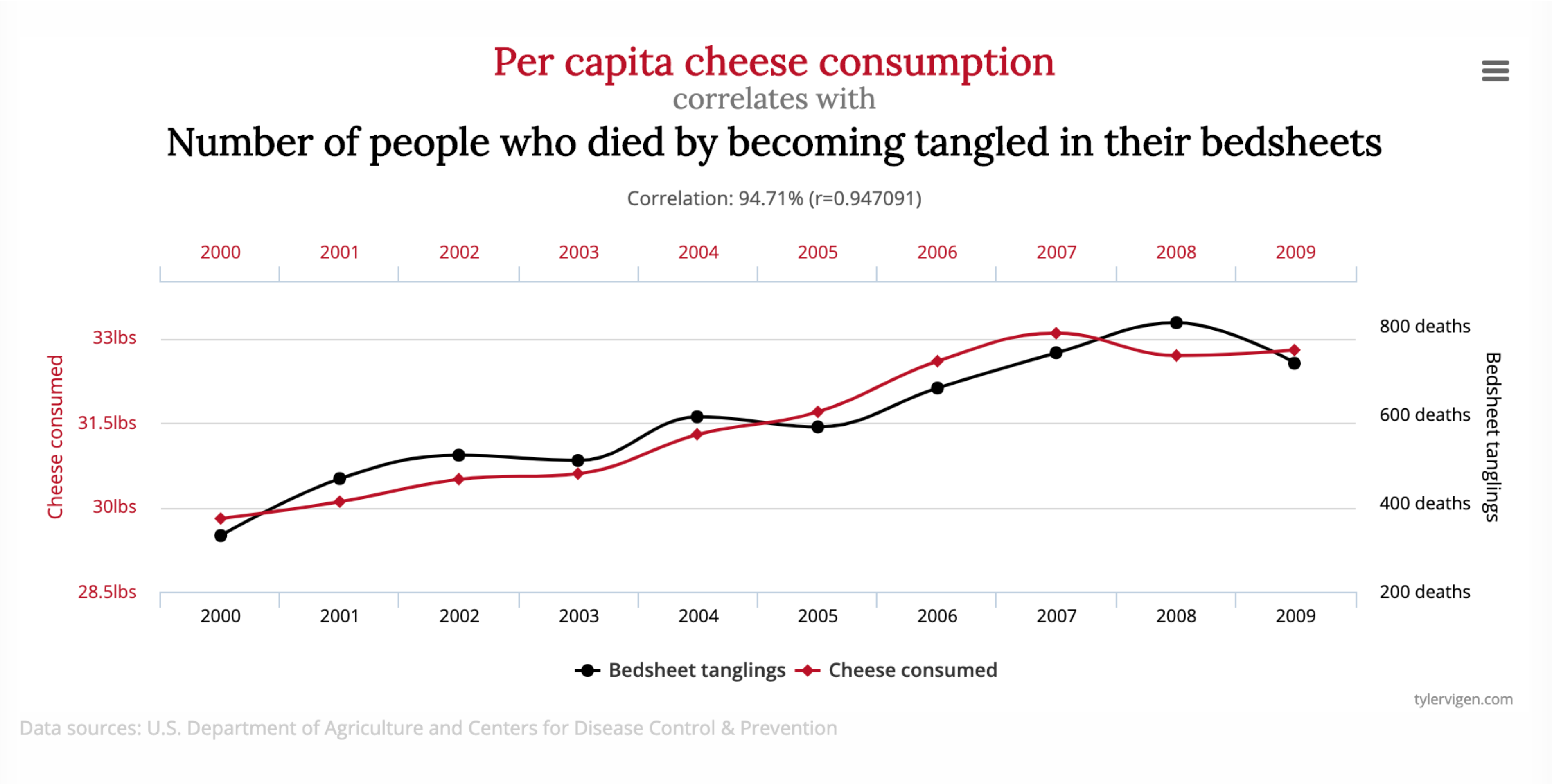
Positive or negative relationship?




The 4 V's of Big Data: volume, velocity, variety, veracity

# “Correlation does not imply causation”




http://tylervigen.com/spurious-correlations

# Useful predictors for time series data

- Trend $y_t = \beta_0 + \beta_1 t + \epsilon_t$ where $t = 1, 2, \dots, T$
- Dummy variables for categorical predictor (later week)
- Trading days
  - # of trading days in a month can vary considerably and can have a substantial effect on sales data.
  - An alternative that allows for the effects of different days of the week has the following predictors

$$\begin{aligned} x_1 &= \text{number of Mondays in month;} \\ x_2 &= \text{number of Tuesdays in month;} \\ &\vdots \\ x_7 &= \text{number of Sundays in month.} \end{aligned}$$

- Distributed lags (later week)

$$\begin{aligned} x_1 &= \text{advertising for previous month;} \\ x_2 &= \text{advertising for two months previously;} \\ &\vdots \\ x_m &= \text{advertising for } m \text{ months previously.} \end{aligned}$$

- Fourier series (later week)

$$x_{1,t} = \sin\left(\frac{2\pi t}{m}\right), x_{2,t} = \cos\left(\frac{2\pi t}{m}\right), x_{3,t} = \sin\left(\frac{4\pi t}{m}\right),$$

$$x_{4,t} = \cos\left(\frac{4\pi t}{m}\right), x_{5,t} = \sin\left(\frac{6\pi t}{m}\right), x_{6,t} = \cos\left(\frac{6\pi t}{m}\right),$$

# Selecting Predictors

1. Adjusted $R^2$, $\bar{R}^2$
   - Widely used; has been around longer than other measures.
   - Maximizing $\bar{R}^2$ is equivalent to minimizing $\hat{\sigma}_e^2$

$$\hat{\sigma}_e^2 = \frac{1}{T-k-1}\sum_{t=1}^{T} e_t^2$$

2. Cross-validation (CV)
   - $T =$ number of observations
   - For large values of $T$, minimizing AIC is equivalent to minimizing CV value (MSE).
3. Information criterion: AIC, AICc, BIC
   - Many statisticians like to use BIC. The textbook uses AICc.
   - If the value of $T$ is large enough, they will all lead to the same model.

# AIC, AICc, BIC

- Akaike's Information Criterion (AIC)
  - The model with the minimum value of the AIC is often the best model for forecasting.
  - AIC penalizes terms more heavily than $\bar{R}^2$
  - Minimizing AIC is asymptotically equivalent to minimizing MSE via leave-one-out cross-validation.
- Corrected Akaike's Information Criterion (AICc)
  - For small values of T, the AIC tends to select too many predictors, and so AICc has been developed.
- Schwarz's Bayesian Information Criterion (BIC)
  - BIC penalizes the number of parameters more heavily than AIC.

$T$ = number of observations
$k$ = number of predictors in the model
SSE = sum of squared error

$$\text{SSE} = \sum_{t=1}^{T} e_t^2$$

$$\text{AIC} = T\log\left(\frac{\text{SSE}}{T}\right) + 2(k+2),$$

AIC = $-2\log(L) + 2(k+2)$
where $L$ is likelihood

$$\text{AIC}_\text{c} = \text{AIC} + \frac{2(k+2)(k+3)}{T-k-3}$$

$$\text{BIC} = T\log\left(\frac{\text{SSE}}{T}\right) + (k+2)\log(T)$$

# Model selection

```
ncs %>%
  mutate(URPR = UR*PR) %>%
  model(
    tslm0 = TSLM(NCS ~ DPIPC),
    tslm1 = TSLM(NCS ~ DPIPC + UR + PR + UMICS),
    tslm2 = TSLM(NCS ~ DPIPC + UR + PR + UMICS + URPR)
    ) %>%
  glance() %>%
  select(.model, adj_r_squared, CV, AIC, AICc, BIC)
```

```
  .model adj_r_squared         CV   AIC  AICc   BIC
  <chr>          <dbl>      <dbl> <dbl> <dbl> <dbl>
1 tslm0          0.215 598194210. 1215. 1215. 1221.
2 tslm1          0.816 153330402. 1131. 1132. 1143.
3 tslm2          0.813 162755872. 1133. 1135. 1147.
```

Based on adjusted $R^2$, we select____

Based on MSE from cross validation, we select ____

Based on AICc, we select _____

# Homework

# Hw: US Personal Consumption and Income

We would like to forecast changes in consumption (y) based on changes in income (x).

Obtain the regression equation

Perform diagnostic test

1. Do signs of coefficients make sense?
2. At significance level $\alpha = 0.05$, is the coefficient of income statistically significant?
3. Explanatory power
4. Residual analysis

```
#################################################
# Homework
#################################################
us_change

help(us_change)
```

```
> us_change
# A tsibble: 198 x 6 [1Q]
  Quarter Consumption Income Production Savings Unemployment
    <qtr>       <dbl>  <dbl>      <dbl>   <dbl>        <dbl>
1 1970 Q1       0.619   1.04     -2.45     5.30          0.9
2 1970 Q2       0.452   1.23     -0.551    7.79          0.5
3 1970 Q3       0.873   1.59     -0.359    7.40          0.5
```

Percentage change

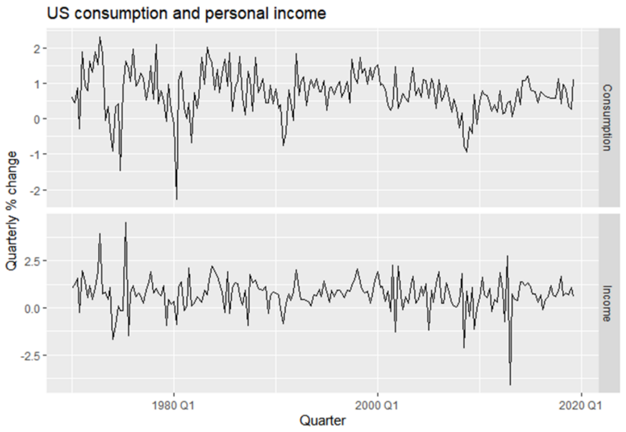
$$y_t = \frac{v_t - v_{t-1}}{v_{t-1}}$$

us_change {fpp3} R Documentation

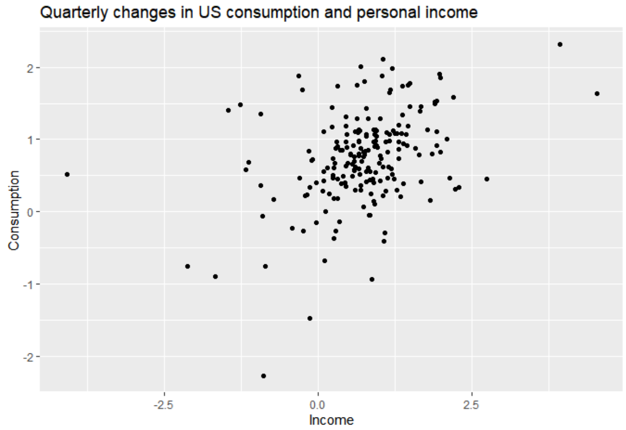
## Percentage changes in economic variables in the USA.

### Description

`us_change` is a quarterly 'tsibble' containing percentage changes in quarterly personal consumption expenditure, personal disposable income, production, savings and the unemployment rate for the US, 1970 to 2016. Original $ values were in chained 2012 US dollars.




```
us_change %>%
  pivot_longer(c(Consumption, Income),
               names_to = "var", values_to = "value") %>%
  ggplot(aes(x = Quarter, y = value)) +
  geom_line() +
  facet_grid(vars(var), scales = "free_y") +
  labs(title = "US consumption and personal income",
       y = "Quarterly % change")
```




```
us_change %>%
  ggplot(aes(x = Income, y = Consumption)) +
  geom_point() +
  labs(x = "Income",
       y = "Consumption",
       title = "Quarterly changes in US consumption and
personal income")
```

# Appendix

Multiple Regression: Matrix Notation

# Regression: Matrix notation

- Recall that multiple regression can be written as $y_t = \beta_0 + \beta_1 x_{1,t} + \cdots + \beta_k x_{k,t} + \epsilon_t$
- Assume $\epsilon_t$ has mean zero and variance $\sigma^2$.
- Let $\boldsymbol{y} = (y_1, \dots, y_n)^\top$, $\boldsymbol{\beta} = (\beta_0, \dots, \beta_k)^\top$, $\boldsymbol{\epsilon} = (\epsilon_1, \dots, \epsilon_n)^\top$.

- The **design (model) matrix**

$$\boldsymbol{X} = \begin{bmatrix} 1 & | & | & & | \\ \vdots & \boldsymbol{x_1} \dots & \boldsymbol{x_i} & \dots & \boldsymbol{x_k} \\ 1 & | & | & & | \end{bmatrix} = \begin{bmatrix} 1 & x_{1,1} & x_{2,1} & \dots & x_{k,1} \\ 1 & x_{1,2} & x_{2,2} & \dots & x_{k,2} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{1,T} & x_{2,T} & \dots & x_{k,T} \end{bmatrix}$$

- $T$ rows reflecting the number of observations
- $k + 1$ columns reflecting the intercept which is represented by the column of ones plus the number of predictors

- Matrix formulation: $\boldsymbol{y} = \boldsymbol{X}\,\boldsymbol{\beta} + \boldsymbol{\epsilon}$

where $\boldsymbol{\epsilon}$ has mean $\mathbf{0}$ and variance $\sigma^2 \boldsymbol{I}$.

- Least squares estimation: Minimize $\boldsymbol{\epsilon}^\top \boldsymbol{\epsilon} = (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})^\top (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})$

# Design (or Model) Matrix
## Example: New Car Sales

- The model matrix

$$\boldsymbol{X} = \begin{bmatrix} 1 & | & | & & | \\ \vdots & \boldsymbol{x}_1 \ldots & \boldsymbol{x}_i & \ldots & \boldsymbol{x}_k \\ 1 & | & | & & | \end{bmatrix}$$

File: `08-TS-Regression`

sheet: `matrix`

| NCS | | ones | DPIPC | UR | PR | UMICS |
|---|---|---|---|---|---|---|
| 152641 | | 1 | 27840 | 5.7 | 4.75 | 93.13 |
| 166036 | | 1 | 28134 | 5.83 | 4.75 | 94.1 |
| 175863 | | 1 | 28168 | 5.73 | 4.75 | 87.27 |
| 151219 | | 1 | 28363 | 5.87 | 4.45 | 83.83 |
| 152843 | | 1 | 28584 | 5.87 | 4.25 | 79.97 |
| 173360 | | 1 | 28958 | 6.13 | 4.24 | 89.27 |
| 178086 | | 1 | 29539 | 6.13 | 4 | 89.3 |
| 154916 | | 1 | 29708 | 5.83 | 4 | 91.97 |

```
dncs <- read_excel("09-TS-regression.xlsx", sheet = "NewCarSales")
X <- model.matrix(NCS ~ DPIPC + UR + PR + UMICS, dncs)
```

```
  (Intercept) DPIPC   UR   PR UMICS
1           1 27840 5.70 4.75 93.13
2           1 28134 5.83 4.75 94.10
3           1 28168 5.73 4.75 87.27
4           1 28363 5.87 4.45 83.83
5           1 28584 5.87 4.25 79.97
6           1 28958 6.13 4.24 89.27
```

# Multiple Regression: Matrix Notation
# Least squares estimates

- Matrix formulation: $\boldsymbol{y} = \boldsymbol{X}\,\boldsymbol{\beta} + \boldsymbol{\epsilon}$
- **Least squares estimation**: Minimize $\boldsymbol{\epsilon}^\top\boldsymbol{\epsilon} = (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})^\top(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})$
- The least squares estimate $\widehat{\boldsymbol{\beta}} = (\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top\boldsymbol{y}$

- This requires the inversion of the matrix $(\boldsymbol{X}^\top\boldsymbol{X})$
- If $X$ is not of full column rank, then matrix $(\boldsymbol{X}^\top\boldsymbol{X})$ is singular, and the model cannot be estimated.

- The residual variance is estimated using

$$\hat{\sigma}_e^2 = \frac{1}{T-k-1}(\boldsymbol{y} - \boldsymbol{X}\hat{\beta})'(\boldsymbol{y} - \boldsymbol{X}\hat{\beta}).$$

# Multiple Regression: Matrix Notation
# Least squares estimate

- Recall that multiple regression can be written as $y_t = \beta_0 + \beta_1 x_{1,t} + \cdots + \beta_k x_{k,t} + \epsilon_t$
- Let $\boldsymbol{y} = (y_1, \dots, y_T)^\top$, $\boldsymbol{\beta} = (\beta_0, \dots, \beta_k)^\top$, $\boldsymbol{\epsilon} = (\epsilon_1, \dots, \epsilon_T)^\top$,
- Matrix formulation: $\boldsymbol{y} = \boldsymbol{X}\,\boldsymbol{\beta} + \boldsymbol{\epsilon}$
- **Least squares estimation**: Minimize $\boldsymbol{\epsilon}^\top \boldsymbol{\epsilon} = (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})^\top (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})$
- The least squares estimate $\widehat{\boldsymbol{\beta}} = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y}$

K2 {=MMULT(MMULT(MINVERSE(MMULT(TRANSPOSE(C2:G61),C2:G61)),TRANSPOSE(C2:G61)),A2:A61)}

| | A | B | C | D | E | F | G | H | I | J | K | L | M |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | NCS | | ones | DPIPC | UR | PR | UMICS | | | | | | |
| 2 | 152641 | | 1 | 27840 | 5.7 | 4.75 | 93.13 | | | beta0 | 25428.26 | | |
| 3 | 166036 | | 1 | 28134 | 5.83 | 4.75 | 94.1 | | | beta1 | 3.227462 | | |
| 4 | 175863 | | 1 | 28168 | 5.73 | 4.75 | 87.27 | | | beta2 | -7666.85 | | |
| 5 | 151219 | | 1 | 28363 | 5.87 | 4.45 | 83.83 | | | beta3 | -3219.5 | | |
| 6 | 152843 | | 1 | 28584 | 5.87 | 4.25 | 79.97 | | | beta4 | 1122.429 | | |
| 7 | 173360 | | 1 | 28958 | 6.13 | 4.24 | 89.27 | | | | | | |
| 8 | 178086 | | 1 | 29539 | 6.13 | 4 | 89.3 | | | | | | |
| 9 | 154916 | | 1 | 29708 | 5.83 | 4 | 91.97 | | | | | | |
| 10 | 159086 | | 1 | 30094 | 5.7 | 4 | 98 | | | | | | |

```
dncs <- read_excel("09-TS-regression.xlsx",
                   sheet = "NewCarSales")
X <- model.matrix(NCS ~ DPIPC + UR + PR + UMICS, dncs)
y <- data.matrix(dncs$NCS)
((solve(t(X) %*% X))%*%t(X)) %*%y
```

```
                   [,1]
(Intercept) 25428.255372
DPIPC           3.227462
UR          -7666.851578
PR          -3219.499125
UMICS        1122.429026
```

# Multiple Regression: Matrix Notation

## Forecasts and prediction intervals

Let $\boldsymbol{x}^*$ be a row vector containing the values of the predictors (in the same format as $\boldsymbol{X}$) for which we want to generate a forecast . Then the forecast is given by

$$\hat{y} = \boldsymbol{x}^*\hat{\boldsymbol{\beta}} = \boldsymbol{x}^*(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{Y}$$

and its estimated variance is given by

$$\hat{\sigma}_e^2\left[1 + \boldsymbol{x}^*(\boldsymbol{X}'\boldsymbol{X})^{-1}(\boldsymbol{x}^*)'\right].$$

A 95% prediction interval can be calculated (assuming normally distributed errors) as

$$\hat{y} \pm 1.96\hat{\sigma}_e\sqrt{1 + \boldsymbol{x}^*(\boldsymbol{X}'\boldsymbol{X})^{-1}(\boldsymbol{x}^*)'}.$$

This takes into account the uncertainty due to the error term $\varepsilon$ and the uncertainty in the coefficient estimates. However, it ignores any errors in $\boldsymbol{x}^*$. Thus, if the future values of the predictors are uncertain, then the prediction interval calculated using this expression will be too narrow.